

x950 Series

Expandable 10G/40G/100G Stackable Layer 3+ Switches



Allied Telesis x950 Series switches are ideal for the modern enterprise network core, where stacking creates a resilient local or distributed solution. These powerful switches support 100 Gigabit connectivity, and the capacity that today's Smart City and IoT networks need.

x950 switches feature 24 or 48 x 1/10G SFP+ ports, or 24 or 48 x 1/2.5/5/10G copper ports, with 4 x 40G/100G uplinks. 24-port models also feature an expansion (XEM) bay to easily add more capacity.

Smart City and IoT networks

Large switching and routing tables support Smart City networks and the Internet of Things (IoT). The x950 Series meets the increasing demand for the convergence of multiple services.

Network automation

Allied Telesis Management Framework™ Plus (AMF Plus) meets the increasing management requirements of modern converged networks, by automating many everyday tasks. AMF Plus has powerful features that allow an entire network to be easily managed as a single virtual device.

Vista Manager™ EX is an intuitive graphical tool for monitoring and managing AMF Plus wired and Autonomous Wave Control (AWC) wireless devices. Full visibility and powerful features enable proactive management of large networks.

Device, network, and security management

The Device GUI on the x950 Series enables graphical monitoring of key switch features to support easy management.

Integrated into the Device GUI, the wireless controller supports visibility and management of AMF Plus wired and AWC wireless network devices, making it ideal as a one-stop solution for small to medium-sized networks. AWC is an intelligent, easy to use Wireless LAN controller that automatically maintains optimal wireless coverage. The wireless controller includes AWC floor and heat maps showing wireless coverage. It also supports AWC Channel Blanket hybrid operation, providing maximum performance and seamless roaming, as well as AWC Smart Connect for simplified deployment, and a resilient Wi-Fi network solution using wireless uplink connectivity.

The AMF-Security mini controller, with management integrated into the Device GUI, ensures protection against internal LAN malware threats, automatically stopping the spread of infection.

Resilient

The convergence of network services has led to increasing demand for highly-available networks with minimal downtime. Virtual Chassis Stacking (VCStack™), in conjunction with link aggregation, provides a network with no single point of failure, and a resilient solution for high-availability applications. The x950 Series can form a VCStack of up to eight units, at any port speed, for enhanced resiliency and simplified management. With VCStack over Long Distance (VCStack LD), stacks can also be created over long distance fiber links, making it the perfect choice for distributed environments too.

Allied Telesis Ethernet Protection Switched Ring (EPSRing™), G.8032 Ethernet Ring Protection, and Media Redundancy Protocol (MRP) ensure that distributed ring-based network segments have resilient access to online resources.

Reliable

Designed with reliability in mind, the x950 Series guarantees the continual delivery of essential services. Hot-swappable components, such as XEMs, fans and load-sharing power supplies, pair with near-hitless online stack reconfiguration to ensure that maintenance doesn't affect network uptime.



Key Features

- High capacity, with 4 x QSFP+/QSFP28 slots supporting 40G or 100G connectivity
- Multi-gig, 10G, 40G, 100G XEMs (28-port models only)
- AC or DC PSU options
- Reverse airflow option for flexible deployment
- Autonomous Management Framework Plus (AMF Plus)
- Large switching and routing tables support Smart City and IoT networks
- VCStack™ up to 8 units, locally or over distance
- EPSRing™ and G.8032 ERPS resilient rings
- EPSR Master
- Media Redundancy Protocol (MRP)
- Active Fiber Monitoring (AFM) for fiber data and stacking links
- Media Access Control Security (MACSec)
- Modbus support
- Multicast Source Discovery Protocol (MSDP)
- Link Monitoring
- Bidirectional Forwarding Detection (BFD)
- Upstream Forwarding Only (UFO)
- VXLAN static tunnels
- AMF-Security mini
- The wireless controller enables:
 - Wired and wireless network visibility
 - AWC wireless network management
 - AWC-Channel Blanket hybrid wireless
 - AWC-Smart Connect wireless uplinks
- Precision Time Protocol (PTP) Transparent Mode
- NETCONF/RESTCONF with YANG data modelling

KEY FEATURES

Wireless controller

Integrated into the Device GUI, the wireless controller provides full network visibility of AMF Plus wired and AWC wireless devices. Manage and simplify wireless deployment with AWC-Smart Connect, and support optimal wireless performance from AWC hybrid operation with maximum throughput and a seamless Wi-Fi user experience.

Autonomous Management Framework Plus (AMF Plus)

AMF Plus is a sophisticated suite of management tools that provide a simplified approach to network management. Common tasks are automated or made so simple that the everyday running of a network can be achieved without the need for highly-trained, and expensive, network engineers. Powerful features like centralized management, auto-backup, auto-upgrade, auto-provisioning and auto-recovery enable plug-and-play networking and zero-touch management.

The x950 Series can operate as the AMF Plus network master, storing firmware and configuration backups for all other network nodes. The AMF Plus master enables auto-provisioning and auto-upgrade by providing appropriate files to new network members.

An AMF Plus license (from AW+ version 5.5.2-2 onwards) provides all standard AMF network management and automation features, and also enables the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10.1 onwards)

AWC Wireless Management

Optimize wireless network performance with the Autonomous Wave Controller (AWC), built-in to the x950 Series. AWC analyzes wireless traffic patterns and automatically reconfigures access points to meet demand.

Wireless network operation in multi-channel, single-channel (Channel Blanket), and hybrid (multi-channel and Channel Blanket) modes, supports maximum data throughput and seamless roaming for the most flexible wireless solution available.

AWC-Smart Connect (AWC-SC) enables plug-and-play wireless network growth, as new APs only need a power connection, and will then automatically create resilient wireless uplink connections to other APs.

Large Network Tables

High-capacity 1.92 Terabit fabric and 1,190Mpps packet forwarding provide powerful data transfer capability, supporting large campus networks as well as Smart City and IoT solutions. Large MAC and IP host tables are ready for the increasing number of connected devices found in modern enterprise and city-wide networks.

Multi-Speed Ports

Copper ports on the x950-28XTm, XEM2-12XTm and XEM2-8XSTm expansion modules support 2.5 and 5 Gigabit connectivity to enable high-speed wireless, or maximum downlink speed using legacy Cat5E/6 cabling.

VCStack™

Create a VCStack of up to eight units at any port speed. VCStack provides a highly available system where network resources are spread out across stacked units, reducing the impact if a unit fails. Aggregating switch ports on different units across the stack provides excellent network resiliency.

VCStack LD

Long-distance stacking allows a VCStack to be created over fiber links to span longer distances, perfect for a distributed network environment.

EPSRing™

EPSRing allows several switches to form protected rings with 50ms failover—perfect for high performance at the core of Enterprise or Provider Access networks. x950 Series switches can act as the EPSR Master.

SuperLoop Protection enables a link between two EPSR nodes to be in separate EPSR domains, improving redundancy and network fault resiliency.

G.8032 Ethernet Ring Protection

G.8032 provides standards-based high-speed ring protection, that can be deployed stand-alone, or interoperate with Allied Telesis EPSR.

Ethernet Connectivity Fault Monitoring (CFM) proactively monitors links and VLANs, and provides alerts when a fault is detected.

Media Recovery Protocol (MRP)

MRP enables high-availability automation networks, and is specified for rings with up to 50 devices, where it guarantees fully deterministic switchover behavior.

NETCONF/RESTCONF

NETCONF/RESTCONF with YANG data modeling provides a standardized way to represent data and securely configure devices.

Active Fiber Monitoring (AFM)

AFM prevents eavesdropping on fiber data or stacking links by monitoring received optical power. If an intrusion is detected, the link can be automatically shut down, or an alert sent.

sFlow

sFlow monitors switched networks, and provides visibility to enable performance optimization, usage accounting/billing, and defense against security threats. Sampled packets sent to a collector provide a real-time view of network traffic.

Media Access Control Security (MAC-Sec)

802.1AE MACSec secures all traffic on point-to-point Ethernet links between directly connected nodes, ensuring protection against security threats such as denial of service, intrusion, man-in-the-middle, passive wiretapping, and playback attacks.

AMF Application Proxy

Allied Telesis SES (Secure Enterprise SDN) solution enables internal LAN threat detection and automatic end-point isolation to protect the network. The AMF Plus Application Proxy enables the SES controller to communicate with the AMF Plus master when a threat is detected, so the AMF Plus master can take action to block the threat at source by quarantining the infected end-point.

Virtual Routing and Forwarding (VRF Lite)

VRF Lite provides Layer 3 network virtualization by dividing a single switch into multiple independent virtual routing domains. With independent routing domains, IP addresses can overlap without causing conflict, allowing multiple customers to have their own secure virtual network within the same physical infrastructure. VRF Lite supports IPv4 and IPv6 unicast and multicast traffic.

The built-in DHCP Server on the x950 Series is VRF aware, enabling the supply of IP addresses to clients across multiple isolated networks.

VLAN Translation

Service Providers can provide customer traffic with a unique VLAN-ID for use within the SP's network.

Enterprises can merge two networks together, without reconfiguring VLAN numbering.

Modbus

Modbus enables communication with Supervisory Control and Data Acquisition (SCADA) systems for industrial automation.

Software-Defined Networking (SDN)

OpenFlow is a key technology that enables the use of SDN to build smart applications that unlock value and reduce cost.

Bi-directional Forwarding Detection (BFD)

BFD enables fast detection of link failures, so recovery time is minimized. BFD works with static routes, and also alongside BGP and OSPF dynamic routing protocols supporting faster shutdown of neighbor connections if a peer session goes down. When using VRF-Lite, BFD is supported globally or within a domain.

Multicast Source Discovery Protocol (MSDP)

MSDP enables two or more PIM-SM (Sparse Mode) domains to share information on active multicast sources, for more efficient forwarding of multicast traffic.

Link Monitoring (Linkmon)

Linkmon enables network health monitoring by regularly sending probes over key links to gather metrics comprising latency, jitter, and probe loss. This supports pro-active network management, and can also be used with triggers to automate a change to device or network configuration in response to the declining health of a monitored link.

AMF-Security mini

The AMF-Sec mini security controller (integrated into the Device GUI) works with your security appliance to enable automatic protection from internal malware threats. It stops the spread of infection and protects the LAN by quarantining any suspect devices.

Flexible deployment

The x950 Series supports both AC and DC PSU options for flexible deployment. Reverse airflow PSUs and fans are also available to support high-speed server connectivity from the top-of-rack in an Enterprise datacenter.

Virtual Extensible LAN (VXLAN) tunnels

VXLAN tunnels let you join two or more L2 networks over an L3 IP network to form a single L2 broadcast domain. VXLAN adds scalability to cloud computing environments. The x950 Series supports static VXLAN tunnels.

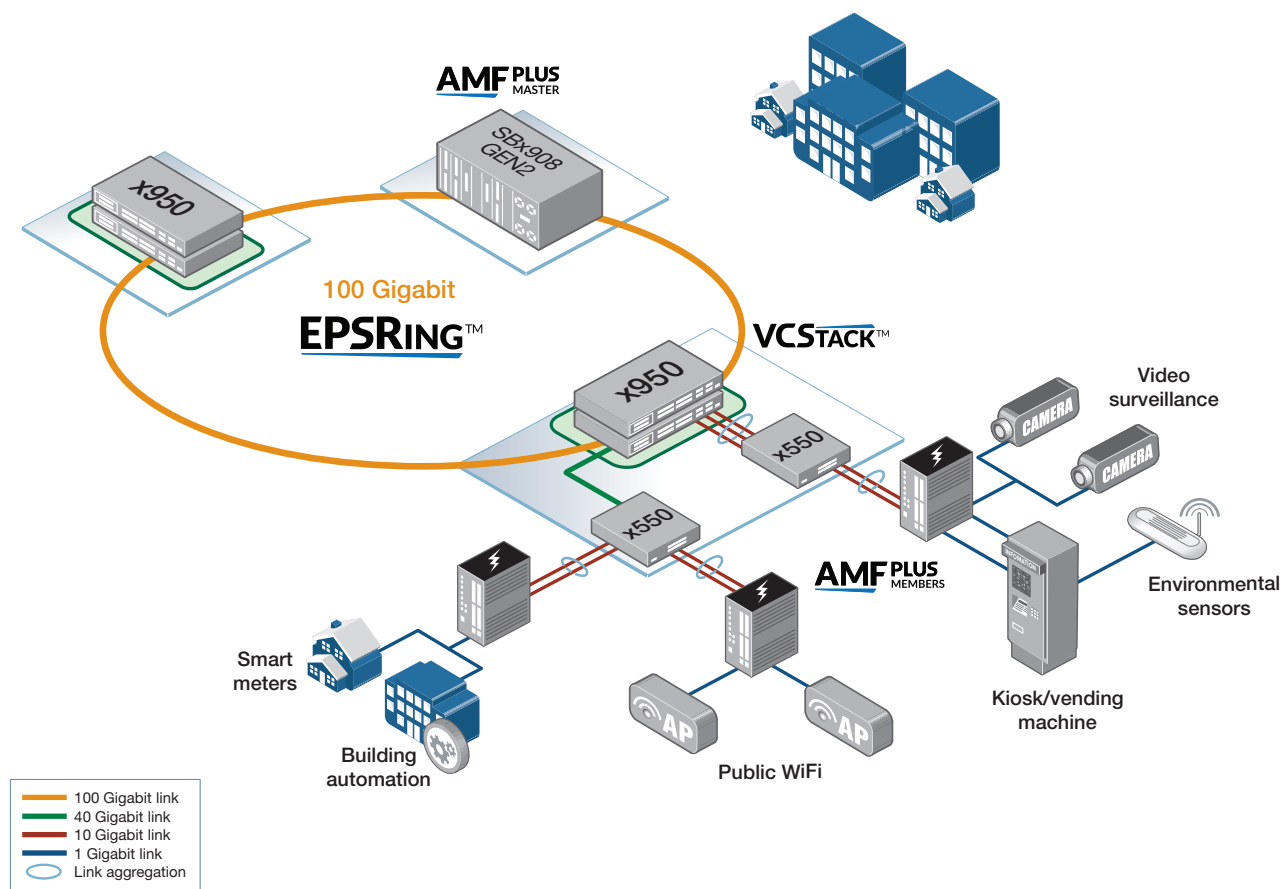
Precision Time Protocol (PTP)

PTP (IEEE 1588v2) synchronizes clocks throughout the network with micro-second accuracy, supporting industrial automation and control systems. PTP operates on standalone or stacked switches.

Distributed network core



Smart city network



All over the world, Smart Cities are looking to increase information availability, security and transport efficiency, whilst reducing pollution and waste. Access to real-time data from a variety of sources gives cities the ability to enhance the quality of their urban services, and increase citizen safety.

In this key solution, x950 Series switches, together with the Allied Telesis SwitchBlade x908 GEN2, create the ideal distributed core solution for Smart City and IoT networks. Large switching and routing tables support the many devices that make up modern metropolitan

networks, including video surveillance cameras, environmental sensors, information kiosks, public Wi-Fi, building automation and many more.

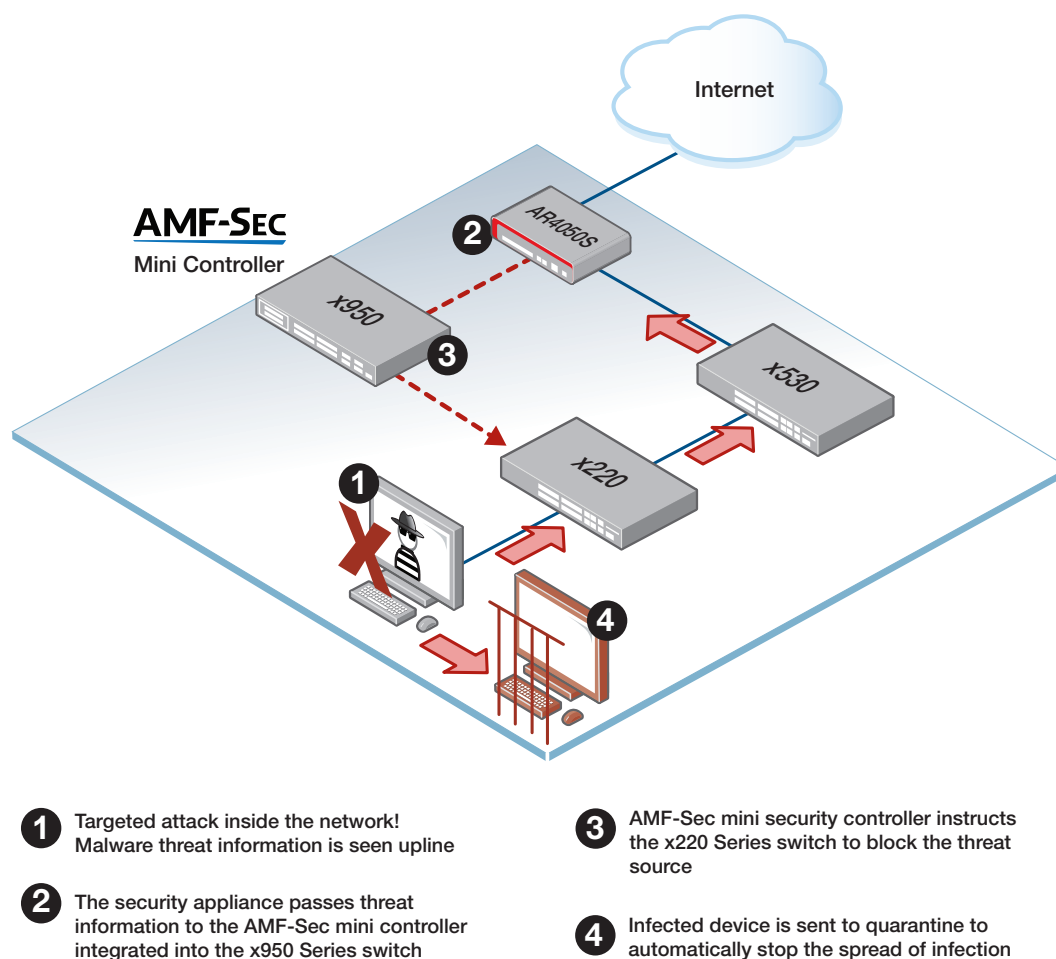
In this Smart City solution, the flexible x950 Series provides 10G, 40G and 100G connectivity. Allied Telesis EPSR creates a high-speed resilient metro ring running at 100Gbps for maximum performance, and extremely fast failover between nodes. EPSR enables rings to recover within as little as 50ms, preventing a node or link failure from impacting the delivery of converged data and video traffic.

AMF Plus automates many day-to-day tasks, backs up the entire network, and provides the ability to configure many or all devices city-wide—with a single command.

The x950 Series and Allied Telesis advanced features enable network managers to deliver leading Smart City services.



Automated Internal LAN Security



AMF Security mini

Most threat protection solutions are only capable of blocking suspicious external traffic arriving at the firewall from the Internet, so only those external threats can be detected and blocked—this is the traditional “secure border” model.

However, the AMF-Sec mini security controller integrated with the x950 Series switch can isolate traffic anywhere in the network, automatically blocking threats such as targeted attacks, or malware introduced inadvertently by staff with USB flash drives, BYOD and so on.

AMF-Sec mini enables automatic protection from internal threats, to protect the LAN from malware by quarantining any suspect devices. Get easy and immediate edge security, so you can relax and enjoy your self-defending network.

The AMF-Sec mini controller can be managed from the Device GUI of the x950 for easy administration. AMF-Sec mini licenses for 1 year or 5 years are available (see the license table in this datasheet).

SPECIFICATIONS

	1/2.5/5/10G (RJ-45) copper ports	1/10 gigabit sfp+ ports	40G/100G QSFP+ /QSFP28 ports	XEM BAY	Switching Fabric	Forwarding Rate
x950-28XSQ		24	4*	1	1.92Tbps	1190Mpps
x950-28XTQm	24		4*	1	1.92Tbps	1190Mpps
x950-52XSQ		48	4*		1.92Tbps	1309Mpps
x950-52XTQm	48		4*		1.92Tbps	1309Mpps

*Can also support up to 16 10G ports (using 4 x 10G breakout cables)

Performance

- Extensive wirespeed traffic classification for ACLs and QoS
- 9KB L2/L3 Jumbo frames
- Wirespeed multicasting
- 96K MAC address entries
- Up to 96K host entries
- Up to 8K multicast entries
- Up to 128 Link Aggregation Groups (LAGS) - any combination of static and dynamic (LACP)
- 4K VLANs (VCStack of up to 4 units)
- 2K VLANs (VCStack of 5-8 units)
- 4GB DDR SDRAM
- 16MB packet buffer memory
- 4GB Flash Memory
- Multicore CPU with 4 dual-threaded cores for high-performance, and enabling wireless network management

Reliability

- Modular AlliedWare Plus operating system
- Dual hot swappable PSUs with 1 + 1 redundancy
- Dual feed support: a separate power circuit can feed each power supply providing extra reliability
- Hot-swappable expansion module (XEM)
- Hot-swappable fan modules
- Full environmental monitoring of PSUs, fans, temperature and internal voltages, with SNMP traps to alert network managers in case of any failure
- MTBF (Mean Time Between Failures)
 - x950-28XSQ: 280,000 hours
 - x950-28XTQm: 190,000 hours
 - x950-52XSQ: 170,000 hours
 - x950-52XTQm: 230,000 hours
 - FAN05: 970,000 hours
 - FAN05R: 970,000 hours
 - PWR600: 280,000 hours
 - PWR600R: 280,000 hours
 - XEM2-12XTm: 900,000 hours
 - XEM2-12XS: 1,370,000 hours
 - XEM2-8XS Tm: 1,070,000 hours
 - XEM2-4QS: 1,840,000 hours
 - XEM2-1CQ: 2,820,000 hours

Expandability

- Support for 4 x 40G or 100G connections built in, and an expansion bay to add further switching capacity
- Versatile licensing options for additional features

Power Characteristics

- AC Voltage: 100 to 240V (+/-10% auto ranging)
- Frequency: 47 to 63Hz

Diagnostic Tools

- Active Fiber Monitoring detects tampering on optical links
- Built-In Self Test (BIST)
- Cable fault locator (TDR)
- Find-me device locator
- Hardware health monitoring
- Automatic link flap detection and port shutdown
- Optical Digital Diagnostic Monitoring (DDM)
- Ping polling for IPv4 and IPv6
- Port mirroring
 - No limit on mirrored ports
 - Up to 4 mirror (analyzer) ports
- VLAN mirroring (RSPAN)
- Link Monitoring
- TraceRoute for IPv4 and IPv6
- Uni-Directional Link Detection (UDLD)

IPv4 Features

- Black hole routing
- Directed broadcast forwarding
- DNS relay
- Equal Cost Multi Path (ECMP) routing
- Policy-based routing
- Route maps
- Route redistribution (OSPF, BGP, RIP)
- Static unicast and multicast routing for IPv4
- UDP broadcast helper (IP helper)
- Up to 600 Virtual Routing and Forwarding (VRF lite) domains (with license)

IPv6 Features

- DHCPv6 client and relay
- DNS client and relay
- IPv4 and IPv6 dual stack
- IPv6 hardware ACLs
- Device management over IPv6 networks with SNMP, Telnet and SSH
- NTP client and server
- Static unicast and multicast routing for IPv6
- Log to IPv6 hosts with Syslog
- IPv6 Ready certified
- VRF Lite

Management

- 7-segment LED provides at-a-glance status and fault information
- Allied Telesis Management Framework Plus (AMF Plus) enables powerful centralized management and zero-touch device installation and recover
- Try AMF Plus for free with the built-in Starter license (includes network management and automation features, but not Vista Manager AMF Plus features)

- NETCONF/RESTCONF northbound interface with YANG data modelling
- Console management port on the front panel for ease of access
- Eco-friendly mode allows ports and LEDs to be disabled to save power
- Web-based Graphical User Interface (GUI)
- Industry-standard CLI with context-sensitive help
- Out-of-band 10/100/1000T Ethernet management port
- Powerful CLI scripting engine
- Comprehensive SNMP MIB support for standards-based device management
- Built-in text editor
- Event-based triggers allow user-defined scripts to be executed upon selected system events
- OpenFlow v1.3 for software-defined network orchestration
- USB interface allows software release files, configurations and other files to be stored for backup and distribution to other devices
- AWC Lite Wi-Fi Controller
- Built-in wireless controller

Quality of Service

- 8 priority queues with a hierarchy of high priority queues for real time traffic, and mixed scheduling, for each switch port
- Bandwidth limiting (virtual bandwidth) Limit bandwidth per port or per traffic class down to 64kbps
- Wirespeed traffic classification with low latency essential for VoIP and real-time streaming media applications
- IPv6 QoS support and IPv6-aware storm protection
- Policy-based QoS based on VLAN, port, MAC and general packet classifiers
- Policy-based storm protection
- Extensive remarking capabilities and taildrop for queue congestion control
- Queue scheduling options for strict priority, weighted round robin or mixed scheduling
- IP precedence and DiffServ marking based on layer 2, 3 and 4 headers

Resiliency Features

- Control Plane Prioritization (CPP) ensures the CPU always has sufficient bandwidth to process network control traffic
- Dynamic link failover (host attach)
- Ethernet Protection Switched Rings (EPSR) with SuperLoop Protection (SLP) and EPSR enhanced recovery for extra resiliency
- G.8032 Ethernet Ring Protection Switching (ERPS)
- Media Redundancy Protocol (MRP)
- Bidirectional Forwarding Detection (BFD)
- Flexi-stacking allows the use of any port speed to stack
- VCStack up to 8 units
- Long-distance VCStack over fiber (VCStack LD)
- Loop protection: loop detection and thrash limiting
- PVST+ compatibility mode
- STP root guard
- VCStack fast failover minimizes network disruption

Security

- Access Control Lists (ACLs) based on layer 3 and 4 headers
- Configurable ACLs for management traffic

- Dynamic ACLs assigned via port authentication

■ ACL Groups enable multiple hosts/ports to be included in a single ACL, reducing configuration

■ Auth fail and guest VLANs

■ Authentication, Authorisation and Accounting (AAA)

■ Bootloader can be password protected for device security

■ BPDU protection

■ DHCP snooping, IP source guard and Dynamic ARP Inspection (DAI)

■ DoS attack blocking and virus throttling

■ Dynamic VLAN assignment

■ MAC address filtering and MAC address lock-down

■ Media Access Control Security (MACSec)

■ Network Access Control (NAC) features manage endpoint security

■ Learn limits (intrusion detection) for single ports or LAGs

■ Private VLANs provide security and port isolation for multiple customers using the same VLAN

■ Simple Certificate Enrollment Protocol (SCEP) supports secure management

■ Secure Copy (SCP)

■ Secure File Transfer Protocol (SFTP) client

■ Strong password security and encryption

■ TACACS+ command authorisation

■ Tri-authentication: MAC-based, web-based and IEEE 802.1x

■ Web-based authentication

■ RADIUS group selection per VLAN or port

■ RADIUS Proxy

Environmental Specifications

- Operating temperature range:
0°C to 50°C (32°F to 122°F)
0°C to 45°C (32°F to 113°F) if using 100G QSFP28 modules
Derated by 1°C per 305 meters (1,000 ft)

■ Storage temperature range:
-25°C to 70°C (-13°F to 158°F)

■ Operating relative humidity range:
5% to 90% non-condensing

■ Storage relative humidity range:
5% to 95% non-condensing

■ Operating altitude:
3,050 meters maximum (10,000 ft)

Electrical Approvals and Compliances

- EMC: EN55032 class A, FCC class A, VCCI class A

■ Immunity: EN55024, EN61000-3-levels 2 (Harmonics), and 3 (Flicker)

Safety

- Standards: UL60950-1, CAN/CSA-C22.2 No. 60950-1-03, EN60950-1, EN60825-1, AS/NZS 60950

■ Certification: UL, cUL, TUV

Restrictions on Hazardous Substances (RoHS) Compliance

- EU RoHS compliant

■ China RoHS compliant

STANDARDS & PROTOCOLS

AlliedWare Plus Operating System

Version 5.5.6-1

Authentication

RFC 1321

MD5 Message-Digest algorithm

RFC 1828

IP authentication using keyed MD5

Border Gateway Protocol (BGP)

BGP dynamic capability

BGP outbound route filtering

RFC 1772

Application of the Border Gateway Protocol (BGP) in the Internet

Power, Heat, Noise (with two PSUs installed)

	Max Power Consumption	Max Heat Dissipation	Noise
x950-28XSQ	231.2W	789.0 BTU/h	63.4 dBA
x950-28XSQ + XEM2-8XSTm*	250.3W	854.0 BTU/h	63.4 dBA
x950-28XSQ + XEM2-12XTm	261.6W	892.8 BTU/h	63.4 dBA
x950-28XSQ + XEM2-12XS v2	262.3W	895.1 BTU/h	63.4 dBA
x950-28XSQ + XEM2-4QS	248.0W	846.4 BTU/h	63.4 dBA
x950-28XSQ + XEM2-1CQ*	238.1W	812.8 BTU/h	63.4 dBA
x950-28XTQm	255.3W	871.1 BTU/h	61.9 dBA
x950-28XTQm + XEM2-8XSTm*	273.9W	934.7 BTU/h	61.9 dBA
x950-28XTQm + XEM2-12XTm	284.6W	971.3 BTU/h	61.9 dBA
x950-28XTQm + XEM2-12XS v2	286.2W	976.6 BTU/h	61.9 dBA
x950-28XTQm + XEM2-4QS	271.7W	927.1 BTU/h	61.9 dBA
x950-28XTQm + XEM2-1CQ*	261.7W	893.2 BTU/h	61.9 dBA
x950-52XSQ	266.1W	908.1 BTU/h	65.7 dBA
x950-52XTQm	300W	1,089 BTU/h	63.5 dBA

Physical Specifications and Latency

	Width	Depth	Height	Mounting	Unpackaged Weight	Packaged Weight	Latency (microseconds)
x950-28XSQ	440 mm (17.32 in)	445 mm (17.52 in)	44 mm (1.73 in)	Rack-mount 1 RU	7.26 kg (16.01 lb)	10.76 kg (23.72 lb)	0.8µs
x950-28XTQm	440 mm (17.32 in)	445 mm (17.52 in)	44 mm (1.73 in)	Rack-mount 1 RU	7.26 kg (16.01 lb)	10.94 kg (24.12 lb)	2.3µs
x950-52XSQ	441 mm (17.36 in)	449 mm (17.68 in)	44 mm (1.73 in)	Rack-mount 1 RU	7.5 kg (16.5 lb)	12.0 kg (26.5 lb)	0.98 µs (100Gbps, FEC)
x950-52XTQm	441 mm (17.36 in)	449 mm (17.68 in)	44 mm (1.73 in)	Rack-mount 1 RU	7.7 kg (16.98 lb)	11.26 kg (24.83 lb)	2.3µs
PWR600-AC	51 mm (2.0 in)	245 mm (9.6 in)	40 mm (1.6 in)	N/A	0.84 kg (1.85 lb)	2.04 kg (4.50 lb)	-
PWR600-DC	51 mm (2.0 in)	245 mm (9.6 in)	40 mm (1.6 in)	N/A	0.84 kg (1.85 lb)	1.84 kg (4.06 lb)	-
FAN05	153 mm (5.99 in)	100 mm (3.94 in)	43 mm (1.69 in)	N/A	0.35 kg (1.85 lb)	1.06 kg (2.34 lb)	-
PWR600R-AC	51 mm (2.0 in)	255 mm (10.04 in)	41 mm (1.62 in)	N/A	0.84 kg (1.85 lb)	2.04 kg (4.50 lb)	-
PWR600R-DC	51 mm (2.0 in)	255 mm (10.04 in)	41 mm (1.62 in)	N/A	0.84 kg (1.85 lb)	2.04 kg (4.50 lb)	-
FAN05R	153 mm (5.99 in)	80 mm (3.15 in)	43 mm (1.69 in)	N/A	0.36 kg (0.79 lb)	1.03 kg (2.27 lb)	-
XEM2-8XSTm*	130 mm (5.11 in)	166 mm (6.53 in)	40 mm (1.57 in)	N/A	0.70 kg (1.54 lb)	1.7 kg (3.75 lb)	2.2µs
XEM2-12XTm	130 mm (5.11 in)	166 mm (6.53 in)	40 mm (1.57 in)	N/A	0.75 kg (1.65 lb)	1.8 kg (3.97 lb)	2.4µs
XEM2-12XS v2	130 mm (5.11 in)	166 mm (6.53 in)	40 mm (1.57 in)	N/A	0.75 kg (1.65 lb)	1.8 kg (3.97 lb)	1.9µs
XEM2-4QS	130 mm (5.11 in)	166 mm (6.53 in)	40 mm (1.57 in)	N/A	0.66 kg (1.45 lb)	1.7 kg (3.75 lb)	0.7µs
XEM2-1CQ*	130 mm (5.11 in)	166 mm (6.53 in)	40 mm (1.57 in)	N/A	0.62 kg (1.37 lb)	1.6 kg (3.53 lb)	0.7µs

*Please contact your sales representative for availability in your region

RFC 1997	BGP communities attribute
RFC 2385	Protection of BGP sessions via the TCP MD5 signature option
RFC 2439	BGP route flap damping
RFC 2545	Use of BGP-4 multiprotocol extensions for IPv6 inter-domain routing
RFC 2858	Multiprotocol extensions for BGP-4
RFC 2918	Route refresh capability for BGP-4
RFC 3392	Capabilities advertisement with BGP-4
RFC 3882	Configuring BGP to block Denial-of-Service (DoS) attacks
RFC 4271	Border Gateway Protocol 4 (BGP-4)
RFC 4360	BGP extended communities
RFC 4456	BGP route reflection - an alternative to full mesh iBGP
RFC 4724	BGP graceful restart
RFC 4893	BGP support for four-octet AS number space
RFC 5065	Autonomous system confederations for BGP

Cryptographic Algorithms

FIPS Approved Algorithms

Encryption (Block Ciphers):

AES (ECB, CBC, CFB and OFB Modes)
3DES (ECB, CBC, CFB and OFB Modes)

Block Cipher Modes:

CCM
CMAC
GCM
XTS

Digital Signatures & Asymmetric Key Generation:

DSA
ECDSA
RSA

Secure Hashing:

SHA-1
SHA-2 (SHA-224, SHA-256, SHA-384, SHA-512)
Message Authentication:
HMAC (SHA-1, SHA-2(224, 256, 384, 512)
Random Number Generation:
DRBG (Hash, HMAC and Counter)

Non FIPS Approved Algorithms

RNG (AES128/192/256)
DES
MD5

Ethernet Standards

IEEE 802.1AE Media Access Control Security (MACSec)
IEEE 802.2 Logical Link Control (LLC)
IEEE 802.3 Ethernet
IEEE 802.3ab1000BASE-T
IEEE 802.3ae10 Gigabit Ethernet
IEEE 802.3an10GBASE-T
IEEE 802.3azEnergy Efficient Ethernet (EEE)
IEEE 802.3ba40GBASE-X
IEEE 802.3bj 100GBASE-X
IEEE 802.3bz2.5GBASE-T and 5GBASE-T
IEEE 802.3x Flow control - full-duplex operation
IEEE 802.3z 1000BASE-X

IPv4 Features

RFC 768	User Datagram Protocol (UDP)
RFC 791	Internet Protocol (IP)
RFC 792	Internet Control Message Protocol (ICMP)
RFC 793	Transmission Control Protocol (TCP)
RFC 826	Address Resolution Protocol (ARP)
RFC 894	Standard for the transmission of IP datagrams over Ethernet networks
RFC 919	Broadcasting Internet datagrams
RFC 922	Broadcasting Internet datagrams in the presence of subnets
RFC 932	Subnetwork addressing scheme
RFC 950	Internet standard subnetting procedure
RFC 951	Bootstrap Protocol (BootP)
RFC 1027	Proxy ARP
RFC 1035	DNS client
RFC 1042	Standard for the transmission of IP datagrams over IEEE 802 networks
RFC 1071	Computing the Internet checksum
RFC 1122	Internet host requirements
RFC 1191	Path MTU discovery
RFC 1256	ICMP router discovery messages
RFC 1518	An architecture for IP address allocation with CIDR

RFC 1519	Classless Inter-Domain Routing (CIDR)
RFC 1542	Clarifications and extensions for BootP
RFC 1591	Domain Name System (DNS)
RFC 1812	Requirements for IPv4 routers
RFC 1918	IP addressing
RFC 2581	TCP congestion control

IPv6 Features

RFC 1981	Path MTU discovery for IPv6
RFC 2460	IPv6 specification
RFC 2464	Transmission of IPv6 packets over Ethernet networks
RFC 2711	IPv6 router alert option
RFC 3484	Default address selection for IPv6
RFC 3587	IPv6 global unicast address format
RFC 3596	DNS extensions to support IPv6
RFC 4007	IPv6 scoped address architecture
RFC 4193	Unique local IPv6 unicast addresses
RFC 4213	Transition mechanisms for IPv6 hosts and routers
RFC 4291	IPv6 addressing architecture
RFC 4443	Internet Control Message Protocol (ICMPv6)
RFC 4861	Neighbor discovery for IPv6
RFC 4862	IPv6 Stateless Address Auto-Configuration (SLAAC)
RFC 5014	IPv6 socket API for source address selection
RFC 5095	Deprecation of type 0 routing headers in IPv6
RFC 5175	IPv6 Router Advertisement (RA) flags option
RFC 6105	IPv6 Router Advertisement (RA) guard

Management

AMF plus MIB and SNMP traps	
AT Enterprise MIB	
Optical DDM MIB	
SNMPv1, v2c and v3	
IEEE 802.1AB Link Layer Discovery Protocol (LLDP)	
RFC 1155	Structure and identification of management information for TCP/IP-based Internets
RFC 1157	Simple Network Management Protocol (SNMP)
RFC 1212	Concise MIB definitions
RFC 1213	MIB for network management of TCP/IP-based Internets: MIB-II
RFC 1215	Convention for defining traps for use with the SNMP
RFC 1227	SNMP MUX protocol and MIB
RFC 1239	Standard MIB
RFC 1724	RIPv2 MIB extension
RFC 2578	Structure of Management Information v2 (SMIv2)
RFC 2579	Textual conventions for SMIv2
RFC 2580	Conformance statements for SMIv2
RFC 2674	Definitions of managed objects for bridges with traffic classes, multicast filtering and VLAN extensions
RFC 2741	Agent extensibility (AgentX) protocol
RFC 2787	Definitions of managed objects for VRRP
RFC 2819	RMON MIB (groups 1,2,3 and 9)
RFC 2863	Interfaces group MIB
RFC 3164	Syslog protocol
RFC 3176	sFlow: a method for monitoring traffic in switched and routed networks
RFC 3411	An architecture for describing SNMP management frameworks
RFC 3412	Message processing and dispatching for the SNMP
RFC 3413	SNMP applications
RFC 3414	User-based Security Model (USM) for SNMPv3
RFC 3415	View-based Access Control Model (VACM) for SNMP
RFC 3416	Version 2 of the protocol operations for the SNMP
RFC 3417	Transport mappings for the SNMP
RFC 3418	MIB for SNMP
RFC 3621	Power over Ethernet (PoE) MIB
RFC 3635	Definitions of managed objects for the Ethernet-like interface types
RFC 3636	IEEE 802.3 MAU MIB
RFC 4022	MIB for the Transmission Control Protocol (TCP)
RFC 4113	MIB for the User Datagram Protocol (UDP)
RFC 4188	Definitions of managed objects for bridges
RFC 4292	IP forwarding table MIB
RFC 4293	MIB for the Internet Protocol (IP)
RFC 4318	Definitions of managed objects for bridges with RSTP

RFC 4560	Definitions of managed objects for remote ping, traceroute and lookup operations
RFC 6527	Definitions of managed objects for VRRPv3

Multicast Support

Bootstrap Router (BSR) mechanism for PIM-SM	
IGMP query solicitation	
IGMP snooping (IGMPv1, v2 and v3)	
IGMP snooping fast-leave	
IGMP/MLD multicast forwarding (IGMP/MLD proxy)	
MLD snooping (MLDv1 and v2)	
PIM for IPv6	
PIM SSM for IPv6	
RFC 1112	Host extensions for IP multicasting (IGMPv1)
RFC 2236	Internet Group Management Protocol v2 (IGMPv2)
RFC 2710	Multicast Listener Discovery (MLD) for IPv6
RFC 2715	Interoperability rules for multicast routing protocols
RFC 3306	Unicast-prefix-based IPv6 multicast addresses
RFC 3376	IGMPv3
RFC 3618	Multicast Source Discovery Protocol (MSDP)
RFC 3810	Multicast Listener Discovery v2 (MLDv2) for IPv6
RFC 3956	Embedding the Rendezvous Point (RP) address in an IPv6 multicast address
RFC 3973	PIM Dense Mode (DM)
RFC 4541	IGMP and MLD snooping switches
RFC 4601	Protocol Independent Multicast - Sparse Mode (PIM-SM): protocol specification (revised)
RFC 4604	Using IGMPv3 and MLDv2 for source-specific multicast
RFC 4607	Source-specific multicast for IP

Open Shortest Path First (OSPF)

OSPF link-local signaling	
OSPF MD5 authentication	
Out-of-band LSDB resync	
RFC 1245	OSPF protocol analysis
RFC 1246	Experience with the OSPF protocol
RFC 1370	Applicability statement for OSPF
RFC 1765	OSPF database overflow
RFC 2328	OSPFv2
RFC 2370	OSPF opaque LSA option
RFC 2740	OSPFv3 for IPv6
RFC 3101	OSPF Not-So-Stubby Area (NSSA) option
RFC 3509	Alternative implementations of OSPF area border routers
RFC 3623	Graceful OSPF restart
RFC 3630	Traffic engineering extensions to OSPF
RFC 4552	Authentication/confidentiality for OSPFv3
RFC 5329	Traffic engineering extensions to OSPFv3
RFC 5340	OSPFv3 for IPv6 (partial support)

Quality of Service (QoS)

IEEE 802.1p	Priority tagging
RFC 2211	Specification of the controlled-load network element service
RFC 2474	DiffServ precedence for eight queues/port
RFC 2475	DiffServ architecture
RFC 2597	DiffServ Assured Forwarding (AF)
RFC 2697	A single-rate three-color marker
RFC 2698	A two-rate three-color marker
RFC 3246	DiffServ Expedited Forwarding (EF)

Resiliency Features

IEC 61439-2 Media Redundancy Protocol (MRP)	
IEEE 802.1AX Link aggregation (static and LACP)	
IEEE 802.1D MAC bridges	
IEEE 802.1s Multiple Spanning Tree Protocol (MSTP)	
IEEE 802.1w Rapid Spanning Tree Protocol (RSTP)	
IEEE 802.3ad Static and dynamic link aggregation	
ITU-T G.8032 Ethernet Ring Protection Switching (ERPS)	
RFC 5798	Virtual Router Redundancy Protocol version 3 (VRRPv3) for IPv4 and IPv6
RFC5880	Bidirectional Forwarding Detection (BFD)

Routing Information Protocol (RIP)

RFC 1058	Routing Information Protocol (RIP)
RFC 2080	RIPng for IPv6
RFC 2081	RIPng protocol applicability statement
RFC 2082	RIP-2 MD5 authentication
RFC 245	3 RIPv2

Security Features

- SSH remote login
- SSLv2 and SSLv3
- TACACS+ Accounting, Authentication and Authorization (AAA)
- IEEE 802.1X Authentication protocols (TLS, TTLS, PEAP and MD5)
- IEEE 802.1X Multi-supplicant authentication
- IEEE 802.1X Port-based network access control
- RFC 2818 HTTP over TLS ("HTTPS")
- RFC 2865 RADIUS authentication
- RFC 2866 RADIUS accounting
- RFC 2868 RADIUS attributes for tunnel protocol support
- RFC 3280 Internet X.509 PKI Certificate and Certificate Revocation List (CRL) profile
- RFC 3546 Transport Layer Security (TLS) extensions
- RFC 3579 RADIUS support for Extensible Authentication Protocol (EAP)
- RFC 3580 IEEE 802.1x RADIUS usage guidelines
- RFC 3748 PPP Extensible Authentication Protocol (EAP)
- RFC 4251 Secure Shell (SSHv2) protocol architecture

- RFC 4252 Secure Shell (SSHv2) authentication protocol
- RFC 4253 Secure Shell (SSHv2) transport layer protocol
- RFC 4254 Secure Shell (SSHv2) connection protocol
- RFC 5176 RADIUS CoA (Change of Authorization)
- RFC 8446 Transport Layer Security (TLS) v1.3
- RFC 8894 Simple Certificate Enrollment Protocol (SCEP)

Services

- RFC 854 Telnet protocol specification
- RFC 855 Telnet option specifications
- RFC 857 Telnet echo option
- RFC 858 Telnet suppress go ahead option
- RFC 1091 Telnet terminal-type option
- RFC 1350 Trivial File Transfer Protocol (TFTP)
- RFC 1985 SMTP service extension
- RFC 2049 MIME
- RFC 2131 DHCPv4 (server, relay and client)
- RFC 2132 DHCP options and BootP vendor extensions
- RFC 2616 Hypertext Transfer Protocol - HTTP/1.1
- RFC 2821 Simple Mail Transfer Protocol (SMTP)
- RFC 2822 Internet message format

- RFC 3046 DHCP relay agent information option (DHCP option 82)
- RFC 3315 DHCPv6 (server, relay and client)
- RFC 3633 IPv6 prefix options for DHCPv6
- RFC 3646 DNS configuration options for DHCPv6
- RFC 3993 Subscriber-ID suboption for DHCP relay agent option
- RFC 4330 Simple Network Time Protocol (SNTP) version 4
- RFC 5905 Network Time Protocol (NTP) version 4

VLAN Support

- Generic VLAN Registration Protocol (GVRP)
- IEEE 802.1ad Provider bridges (VLAN stacking, Q-in-Q)
- IEEE 802.1Q Virtual LAN (VLAN) bridges
- IEEE 802.1v VLAN classification by protocol and port
- IEEE 802.3ac VLAN tagging
- RFC 7348 VxLAN (Static Tunnels only)

Voice over IP (VoIP)

- LLDP-MED ANSI/TIA-1057
- Voice VLAN

Feature Licenses

	Description	Includes	Stack Licensing
AT-FL-x950-01	x950 Premium license	■ OSPF¹ (16,000 routes) ■ BGP4¹ (5,000 routes) ■ PIMv4-SM, DM and SSM (2,000 entries) ■ VLAN double tagging (Q-in-Q) ■ RIPng (5,000 routes) ■ OSPFv3 (8,000 routes) ■ BGP4+ (5,000 routes) ■ MLDv1 and v2 ■ PIMv6-SM and SSM (1,000 entries) ■ VRF lite (64 domains) ■ RADIUS Full ■ UDLD ■ VLAN Translation ■ VXLAN ■ PTP Transparent Mode	■ One license per stack member
AT-SW-APM10-1YR ^{2,3}	Cumulative AMF Plus Master license	■ AMF Plus Master license for up to 10 nodes for 1 year	■ One license per stack
AT-SW-APM10-5YR ^{2,3}	Cumulative AMF Plus Master license	■ AMF Plus Master license for up to 10 nodes for 5 years	■ One license per stack
AT-FL-x950-AAP-1YR	AMF Application Proxy license	■ AMF Application Proxy license for 1 year	■ One license per stack
AT-FL-x950-AAP-5YR	AMF Application Proxy license	■ AMF Application Proxy license for 5 years	■ One license per stack
AT-FL-x950-OF13-1YR	OpenFlow license	■ OpenFlow v1.3 for 1 year	■ Not supported on a stack
AT-FL-x950-OF13-5YR	OpenFlow license	■ OpenFlow v1.3 for 5 years	■ Not supported on a stack
AT-FL-x950-8032	ITU-T G.8032 license	■ G.8032 ring protection ■ Ethernet CFM	■ One license per stack member
AT-FL-x950-MODB	Modbus license	■ Modbus for industrial applications	■ One license per stack member
AT-FL-x950-MSEC ⁴	MACSec license	■ Media Access Control Security	■ One license per stack member
AT-FL-x950-VLF	VRF-Lite Full license	■ VRF lite (600 domains)	■ One license per stack member
AT-FL-x950-ASEC-1YR	AMF-Sec license	■ AMF-Sec license for 1 year	■ One license per stack
AT-FL-x950-ASEC-5YR	AMF-Sec license	■ AMF-Sec license for 5 years	■ One license per stack
AT-SW-AWC10-1YR ⁵	Cumulative AWC license	■ Autonomous Wave Control (AWC) license for up to 10 access points for 1 year	■ One license per stack
AT-SW-AWC10-5YR ⁵	Cumulative AWC license	■ Autonomous Wave Control (AWC) license for up to 10 access points for 5 years	■ One license per stack
AT-SW-CB10-1YR-2022 ⁶	Cumulative AWC-CB and AWC-SC license	■ AWC Channel Blanket and AWC Smart Connect license for up to 10 access points for 1 year	■ One license per stack
AT-SW-CB10-5YR-2022 ⁶	Cumulative AWC-CB and AWC-SC license	■ AWC Channel Blanket and AWC Smart Connect license for up to 10 access points for 5 years	■ One license per stack

¹ 64 OSPF and BGP routes included in base license

² From AW+ version 5.5.2-2 onwards, AMF Plus licenses provide all standard AMF network management and automation features. They also enable the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10.1 onwards)

³ Purchase one license per 10 nodes (up to 180 nodes maximum)

⁴ MACSec only operates on the XEM2-12XS v2 and XEM2-8XSTm expansion modules

⁵ 5 APs can be managed for free. Purchase one license per 10 additional APs (up to 180 APs maximum)

⁶ Channel Blanket and Smart Connect are not available as free services. Both an AWC-CB license and an AWC license are required for Channel Blanket and/or Smart Connect to operate. Purchase one AWC-CB license per 10 APs (up to 180 APs maximum).

ORDERING INFORMATION

AT-x950-28XSQ-BOy^{7,8,9}	24-port 1/10G SFP/SFP+ stackable switch with 4 x 40G/100G QSFP+/QSFP28 ports, a XEM bay, and dual hotswap PSU and Fan bays
AT-x950-28XTQm-BOy^{7,8,9}	24-port 1/2.5/5/10G copper stackable switch with 4 x 40G/100G QSFP+/QSFP28 ports, a XEM bay, and dual hotswap PSU and Fan bays
AT-x950-52XSQ-BOy^{7,8,9}	48-port 1/10G SFP/SFP+ stackable switch with 4 x 40G/100G QSFP+/QSFP28 ports, and dual hotswap PSU and Fan bays
AT-x950-52XTQm-BOy^{7,8}	48-port 100M(FD)/1G/2.5G/5G/10G copper ports with 4 x 40G/100G ports QSFP+/QSFP28 ports, and dual hotswap PSU and Fan bays
AT-x950-28XSQ^{8,9}	24-port 1/10G SFP/SFP+ stackable switch with 4 x 40G/100G QSFP+/QSFP28 ports, a XEM bay, and dual hotswap PSU and Fan bays
AT-x950-28XTQm^{8,9}	24-port 1/2.5/5/10G copper stackable switch with 4 x 40G/100G QSFP+/QSFP28 ports, a XEM bay, and dual hotswap PSU and Fan bays
AT-x950-52XSQ^{8,9}	48-port 1/10G SFP/SFP+ stackable switch with 4 x 40G/100G QSFP+/QSFP28 ports, and dual hotswap PSU and Fan bays
AT-x950-52XTQm⁸	48-port 100M(FD)/1G/2.5G/5G/10G copper ports with 4 x 40G/100G ports QSFP+/QSFP28 ports, and dual hotswap PSU and Fan bays
AT-RKMT-SL01	Sliding rack mount kit
AT-FAN05	Spare hot-swappable fan module
AT-PWR600-80⁸	600W DC system power supply
AT-PWR600-xx^{8,10}	600W AC system power supply
AT-FAN05R	Hot-swappable fan module (reverse airflow)
AT-PWR600R-80⁸	600W DC system power supply (reverse airflow)
AT-PWR600R-xx^{8,10}	600W AC system power supply (reverse airflow)
AT-XEM2-8XSTm*	4 x 1/2.5/5/10G RJ45 ports and 4 x 1G/10G SFP+ ports
AT-XEM2-12XTm	12 x 1/2.5/5/10G RJ45 ports
AT-XEM2-12XS v2	12 x 1G/10G SFP+ ports
AT-XEM2-4QS	4 x 40G QSFP+ ports
AT-XEM2-1CQ*	1 x 100G QSFP28 port

⁷ Where Oy = 01 for 1 year Net Cover support
05 for 5 years Net Cover support

⁸ Note that fans are included but NO power supplies ship with the base chassis, they must be ordered separately

* Please contact your sales representative for availability in your region

⁹ Trade Act Agreement compliant

¹⁰ Where xx = 10 for US power cord
30 for UK power cord
40 for AU power cord
50 for EU power cord

Accessories

100G QSFP28 Modules	
AT-QSFP28SR4	100GSR 850 nm short-haul up to 100 m with MMF
AT-QSFP28LR4	100GLR 1310 nm medium-haul, 10 km with SMF
AT-QSFP28ZR4	100GZR 1310 nm long-haul, 80 km with SMF, TAA ⁹
AT-QSFP28-1CU	1 meter QSFP28 direct attach cable
AT-QSFP28-3CU	3 meter QSFP28 direct attach cable
40G QSFP+ Modules	
AT-QSFP1CU	1 meter QSFP+ direct attach cable
AT-QSFP3CU	3 meter QSFP+ direct attach cable
AT-QSFPSR4	40GSR4 850nm short-haul up to 150 m with MMF
AT-QSFPSR4LCa	40GSR4 850 nm short-haul up to 150 m with MMF, LC
AT-QSFPLR4	40GLR4 1310 nm medium-haul, 10 km with SMF
AT-QSFPER4	40GER4 1310 nm long-haul, 40 km with SMF
Breakout Cables For 4 x 10G connections	
AT-QSFP-4SFP10G-3CU	QSFP to 4 x SFP+ breakout direct attach cable (3m)
AT-QSFP-4SFP10G-5CU	QSFP to 4 x SFP+ breakout direct attach cable (5m)
10GbE SFP+ Modules	
AT-SP10SR	10GSR 850 nm short-haul, 300 m with MMF
AT-SP10SR/I	10GSR 850 nm short-haul, 300 m with MMF industrial temperature
AT-SP10LRa/I	10GLR 1310 nm medium-haul, 10 km with SMF industrial temperature
AT-SP10ER40a/I	10GER 1310 nm long-haul, 40 km with SMF industrial temperature
AT-SP10BD10/I-12	10 GbE Bi-Di (1270 nm Tx, 1330 nm Rx) fiber up to 10 km industrial temperature, TAA ⁹
AT-SP10BD10/I-13	10 GbE Bi-Di (1330 nm Tx, 1270 nm Rx) fiber up to 10 km industrial temperature, TAA ⁹
AT-SP10BD20-12	10 GbE Bi-Di (1270 nm Tx, 1330 nm Rx) fiber up to 20 km, TAA ⁹
AT-SP10BD20-13	10 GbE Bi-Di (1330 nm Tx, 1270 nm Rx) fiber up to 20 km, TAA ⁹

AT-SP10BD20-12	10 GbE Bi-Di (1270 nm Tx, 1330 nm Rx) fiber up to 20 km, TAA ⁹
AT-SP10BD20-13	10 GbE Bi-Di (1330 nm Tx, 1270 nm Rx) fiber up to 20 km, TAA ⁹
AT-SP10BD40/I-12	10 GbE Bi-Di (1270 nm Tx, 1330 nm Rx) fiber up to 40 km industrial temperature, TAA ⁹
AT-SP10BD40/I-13	10 GbE Bi-Di (1330 nm Tx, 1270 nm Rx) fiber up to 40 km industrial temperature, TAA ⁹
AT-SP10TM	1G/2.5G/5G/10G, 100 m copper, TAA ⁹
10GbE SFP+ Cables	
AT-SP10TW1	1 meter SFP+ direct attach cable
AT-SP10TW3	3 meter SFP+ direct attach cable
AT-SP10TW7	7 meter SFP+ direct attach cable
1000Mbps SFP Modules	
AT-SPSX/I	1000SX GbE multi-mode 850 nm fiber up to 550 m industrial temperature, TAA ⁹
AT-SPTXc	1000T 100 m copper
AT-SPSX	1000SX GbE multi-mode 850 nm fiber up to 550 m
AT-SPEX	1000X GbE multi-mode 1310 nm fiber up to 2 km
AT-SPLX10a	1000LX GbE single-mode 1310 nm fiber up to 10 km
AT-SPLX10a/I	1000LX GbE single-mode 1310 nm fiber up to 10 km industrial temperature
AT-SPBD10-13	1000LX GbE Bi-Di (1310 nm Tx, 1490 nm Rx) fiber up to 10 km
AT-SPBD10-14	1000LX GbE Bi-Di (1490 nm Tx, 1310 nm Rx) fiber up to 10 km
AT-SPBD20-13/I	1000BX GbE Bi-Di (1310 nm Tx, 1490 nm Rx) fiber up to 20 km industrial temperature
AT-SPBD20-14/I	1000BX GbE Bi-Di (1490 nm Tx, 1310 nm Rx) fiber up to 20 km industrial temperature
AT-SPBD40-13/I	1000LX GbE single-mode Bi-Di (1310 nm Tx, 1490 nm Rx) fiber up to 40 km, industrial temperature

⁹ Trade Act Agreement compliant

1000Mbps SFP Modules	
AT-SPBD40-14/I	1000LX GbE single-mode Bi-Di (1490 nm Tx, 1310 nm Rx) fiber up to 40 km, industrial temperature
AT-SPLX40	1000LX GbE single-mode 1310 nm fiber up to 40 km
AT-SPZX80	1000ZX GbE single-mode 1550 nm fiber up to 80 km
AT-SPZX120/I	1000ZX GbE single-mode 1550 nm fiber up to 120 km industrial temperature